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Clinical Impact of Rapid Molecular Diagnosis of Bloodstream Infections: A Randomized Controlled Trial

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Background. Rapid identification of pathogens and antimicrobial resistance is important for optimizing therapies for bloodstream infections. We conducted a randomized controlled trial comparing standard of care (SOC) culture and phenotypic antimicrobial susceptibility testing with SOC plus BioFire BCID2 testing (Multiplex PCR arm).

Methods. Patients with positive blood cultures were randomized 1:1 to the SOC or multiplex PCR arm, both with antimicrobial stewardship. The primary outcome was the time from randomization to the first effective antimicrobial modification within 120 hours. The secondary outcomes were persistent bloodstream infections, mortality, and length of hospital stay.

Results. Among 418 patients (208 SOC, 210 multiplex PCR), the median time to first effective antimicrobial modification was significantly shorter in the multiplex PCR arm (23.2 vs 64.5 hours; $P < .001$). This was consistent across Gram-negative (30.5 vs 77.1 hour; $P < .001$), Gram-positive (22.7 vs 53.5 hours; $P = .001$), and yeast infections (2.9 vs 27.7 hours; $P < .001$). The reduction was maintained regardless of laboratory hours (regular hours: 9.3 vs 66.7 hours; off-hours: 37.5 vs 60.5 hours; $P < .001$) and intensive care unit (ICU) status (ICU: 20.5 vs 84.8 hours; $P < .001$; non-ICU: 23.4 vs 55.5 hours; $P < .001$). The multiplex PCR arm showed a lower rate of persistent bloodstream infections (5.7% vs 11.5%, $P = .038$). No significant differences were observed in 30-day mortality or length of stay.

Conclusions. Rapid multiplex PCR shortened the time to effective antimicrobial modification in patients with bloodstream infections.

Keywords. blood cultures; rapid diagnostics; bloodstream infection; PCR; antimicrobial resistance.

Bloodstream infections are major causes of morbidity and mortality worldwide [1]. Early identification of the causative pathogen and timely optimization of antimicrobial therapy are essential for improving patient outcomes [2]. However, conventional blood culture-based workflows require subculture after a positive blood culture for organism identification and antimicrobial susceptibility testing (AST), which can delay the optimization of antimicrobial therapy [2]. Novel platforms are available for rapid pathogen identification and AST of blood cultures, which can improve the treatment and outcomes of bacteremia [3]. Rapid multiplex molecular diagnostic platforms identify bloodstream pathogens and selected resistance genes from positive blood cultures, substantially shortening the time required to obtain microbiological results [4]. In addition, a novel rapid fluorescence in situ hybridization-based identification method combined with microfluidics-based phenotypic AST results in faster changes in antibiotic therapy for bloodstream infections [5]. Early availability of phenotypic antimicrobial susceptibility information can enhance antimicrobial stewardship (AS) efforts [6]. More recently, a multicenter, multinational randomized controlled trial (RCT) demonstrated faster antibiotic escalation or de-escalation with rapid phenotypic AST [7]. Accordingly, integrating rapid diagnostic testing with AS programs may be an effective strategy for optimizing antimicrobial therapy [8].

Despite the growing availability of rapid diagnostic platforms that provide comprehensive pathogen identification and antimicrobial resistance information, high-quality evidence from RCTs remains limited. To the best of our knowledge, only 2 prospective RCTs have evaluated the clinical impact of rapid identification of Gram-negative bacteria, Gram-positive bacteria, and yeasts, together with resistance marker detection, directly from positive blood cultures [2, 9]. These studies demonstrated potential benefits in optimizing antimicrobial therapy. However, further evidence is needed to establish the clinical impact of implementing such approaches across different healthcare settings and stewardship

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environments. Therefore, we conducted a prospective RCT to determine whether rapid multiplex PCR for direct pathogen identification and resistance marker detection from positive blood cultures improves antimicrobial management and clinical outcomes in patients with bloodstream infections.

METHODS

Study Design

This prospective, open-label, single-center RCT was conducted at the Korea University Anam Hospital. The study was conducted between June and October 2025.

Participants

Hospitalized patients who had positive blood cultures were screened for eligibility. The clinical microbiology laboratory operated with full staffing from Monday to Saturday, 7:00–16:00 (regular laboratory hours). Eligible participants were adults (≥18 years) with positive blood cultures who were randomized within 24 hours of blood culture positivity. Patients were excluded if they had previously been enrolled, had a known positive blood culture within the preceding 7 days, had non-blood specimens inoculated into blood culture bottles at randomization, or died or were transferred to another hospital before randomization. Patients for whom BCID2 testing had already been ordered before randomization were excluded when the prespecified allocation was to the standard of care (SOC) group. Patients assigned to the SOC group were also excluded if BCID2 testing was subsequently ordered as part of routine clinical care after randomization. The study protocol was approved by the Institutional Review Board of the Korea University Anam Hospital, who waived the requirement for written informed consent.

Randomization and Blinding

Patients were randomized to either the SOC or multiplex PCR group in a 1:1 ratio using 4 permuted blocks. Randomization was performed following the identification of a positive blood culture using an automated system. Owing to the diagnostic nature of the study, treatment providers were not blinded to the group assignments, and as the molecular results were integrated directly into the electronic medical records for real-time reporting, clinicians had immediate access to the results to make timely and informed decisions regarding antimicrobial modification.

Laboratory Tests and Reporting

All blood cultures were processed using the BACT/ALERT VIRTUO system (bioMérieux). Upon positivity, subculture and Gram stain were performed, followed by MALDI-TOF MS identification (Bruker Sirius; Bruker Daltonics, Bremen, Germany) after overnight incubation. Phenotypic AST was performed using VITEK 2 (bioMérieux) or MicroScan (Beckman Coulter, Brea, CA, United States). Carbapenem-resistant *Enterobacterales* were tested for carbapenemase production using NG-Test CARBA 5 (NG Biotech, Guipry, France). This workflow constituted the SOC. Patients randomized to the multiplex PCR group additionally underwent BioFire BCID2 Panel testing (bioMérieux, Marcy-l'Étoile, France), which detects 15 Gram-negative bacteria, 8 Gram-positive bacteria, 7 yeasts, and 10 antimicrobial resistance genes directly from positive blood cultures [10, 11]. During regular laboratory hours, all procedures, including BCID2 testing, were performed. During off-hours, only subculture was performed, with all remaining procedures completed during the next regular laboratory shift. Results were reported through the electronic medical record as they became available.

Antimicrobial Stewardship

In addition to routine reporting through electronic medical records, the laboratory notified the treating providers via text messages after randomization, recommending an infectious disease (ID) consultation for appropriate antimicrobial management. The laboratory also notified the treating providers of the initial organism identification for each group by text messages. Decisions to request an ID consultation and accept ID recommendations were left to the discretion of the treating providers. Antimicrobial use of the selected agents was routinely monitored as part of an institutional AS program. This approach reflects routine clinical practice, rather than study-specific AS interventions.

Outcomes

The primary outcome was the time from randomization to the first effective antimicrobial modification, defined as either escalation, de-escalation, or lateral change within 120 hours. The earliest documented modification of administered antimicrobial therapy was recorded as the time to modification for all patients, including those who died. Effective treatment was defined as administered antimicrobials with in vitro activity based on phenotypic minimum inhibitory concentration results. Patients without treatment modification within 120 hours were assigned a value of 120 hours, and no additional censoring was applied. Antimicrobial de-escalation was defined as a reduction in the number of antibiotics, a change to a narrower-spectrum agent based on the antibiotic ranking schema [5, 12], or a switch from intravenous to oral therapy [5]. Antimicrobial escalation was defined as an increase in the number or rank of agents [5, 12] or a switch from oral to intravenous therapy [5]. For patients with yeast infections, antifungal de-escalation was defined as a change from echinocandins to azoles or the discontinuation of antifungal therapy, whereas antifungal escalation was defined as a change from azoles to echinocandins or the addition of antifungal therapy [13]. Changes not classified as escalation or de-escalation,

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mainly reflecting movements within the same rank despite the application of three predefined references [5, 12, 13], were defined as lateral changes. The primary outcome was assessed across prespecified subgroups, defined by the timing of blood culture positivity (regular laboratory hours vs off-hours), organism type (Gram-negative, Gram-positive, or yeast), patient location (intensive care unit [ICU] vs non-ICU), antimicrobial resistance (with vs without resistance), and antimicrobial modifications (excluding cases with no change). A post hoc exploratory subgroup analysis was performed according to BCID2 on-panel versus off-panel organisms. Antimicrobial resistant bacteria included extended-spectrum β-lactamase (ESBL)–producing *Enterobacterales*, carbapenem-resistant *Enterobacterales* (CRE), multidrug-resistant *Pseudomonas* spp., multidrug-resistant *Acinetobacter* spp., oxacillin-resistant *Staphylococcus* spp., vancomycin-resistant *Enterococcus* spp., and multidrug-resistant *Candida* spp. Multidrug resistance was defined as nonsusceptibility to ≥1 agent in ≥3 antimicrobial categories for *Acinetobacter* spp. and *Pseudomonas* spp. [14] and to ≥1 agent in ≥2 antifungal classes for *Candida* spp. [15].

Secondary outcomes included microbiological outcomes (persistent bloodstream infection, defined as the isolation of the same organism from follow-up blood cultures obtained ≥3 days after randomization for bacterial infections and ≥5 days for fungal infections, and acquisition of organisms with antimicrobial resistance, *Clostridioides difficile*, or candidemia, as defined in the Supplementary Appendix), ID consultation rates, and clinical outcomes, including the proportion of patients with effective antimicrobial modifications at 12, 24, 48, 72, 96, and 120 hours after randomization; 30-day mortality; length of hospital stay; length of ICU stay; ICU admission rates for patients initially admitted to non-ICU wards; discharge within 30 days of randomization; readmission within 30 days after discharge; and hospital costs. Blood culture contamination was defined as the growth of common commensal organisms in a single blood culture set when 2 or more sets were obtained [16]. Additional methodological details are provided in the Supplementary Methods.

Statistical Analyses

The primary analysis used a modified intention-to-treat (mITT) approach including all randomized patients with minimal post-randomization exclusions based on predefined criteria. Categorical variables were compared using Fisher's exact test and continuous variables using the Mann–Whitney U test after the Shapiro–Wilk normality test. Cumulative event curves for antimicrobial modification were generated using the Kaplan–Meier method. A total of 428 patients provided 80% power to detect a 23-hour difference in time to first antimicrobial modification between groups (2-sided $\alpha = 0.05$), assuming a standard deviation of 48.5 hours based on our previous study [4] and accounting for right-skewing due to patients without antimicrobial modification within 120 hours. Analyses were performed using GraphPad Prism version 11.0.0 (GraphPad Software, San Diego, CA, United States). $P < .05$ was considered statistically significant.

RESULTS

Study Participants

A total of 428 positive blood culture cases were screened and unique patients were randomized; 418 were included in the mITT population (Figure 1). The baseline sociodemographic and clinical characteristics were comparable between the 2 study arms (Table 1). Community-onset infections accounted for less than half of the cases in both arms. The most common source of bacteremia was intra-abdominal infection.

The distribution of pathogens and antimicrobial resistance profiles was comparable between the study arms (Table 2). Polymicrobial infections occurred in 6.7% and 5.7% of patients in the SOC and multiplex PCR arms, respectively. Overall, 222 organisms were identified in 208 patients in the SOC arm and 216 organisms in 210 patients in the multiplex PCR arm; 87.4% and 90.3%, respectively, were BCID2 on-panel organisms. The median time from blood culture positivity to randomization was similar between the SOC and multiplex PCR arms (5.0 vs 4.2 hours, $P = .566$). The median time from randomization to the BCID2 result was 1.9 hours, whereas the times to MALDI-TOF MS identification and phenotypic AST were comparable between the groups (Table 3).

Primary Outcome

The time from randomization to the first effective antimicrobial modification was significantly shorter in the multiplex PCR arm than in the SOC arm (23.2 vs 64.5 hours, $P < .001$) (Table 4 and Figure 2*A*). The median number of hours was consistently shorter in the multiplex PCR arm across all prespecified subgroups, according to laboratory working hours (regular vs off-hours), pathogen type, ICU status, antimicrobial resistance status, and antibiotic escalation or de-escalation, as well as among patients with on-panel pathogens; however, this difference was not observed among those with off-panel organisms (Table 4 and Figure 2).

Secondary Outcomes

The proportions of antimicrobial escalation, de-escalation, lateral change, and no modification were comparable between the study arms (Table 5). Notably, the absence of effective therapy was more frequent in the SOC arm (Table 5). Accordingly, the proportion of patients receiving effective therapy was consistently higher in the multiplex PCR arm at all evaluated time points (Table 6). The causative organisms and reasons for ineffective therapy among patients without effective antimicrobial

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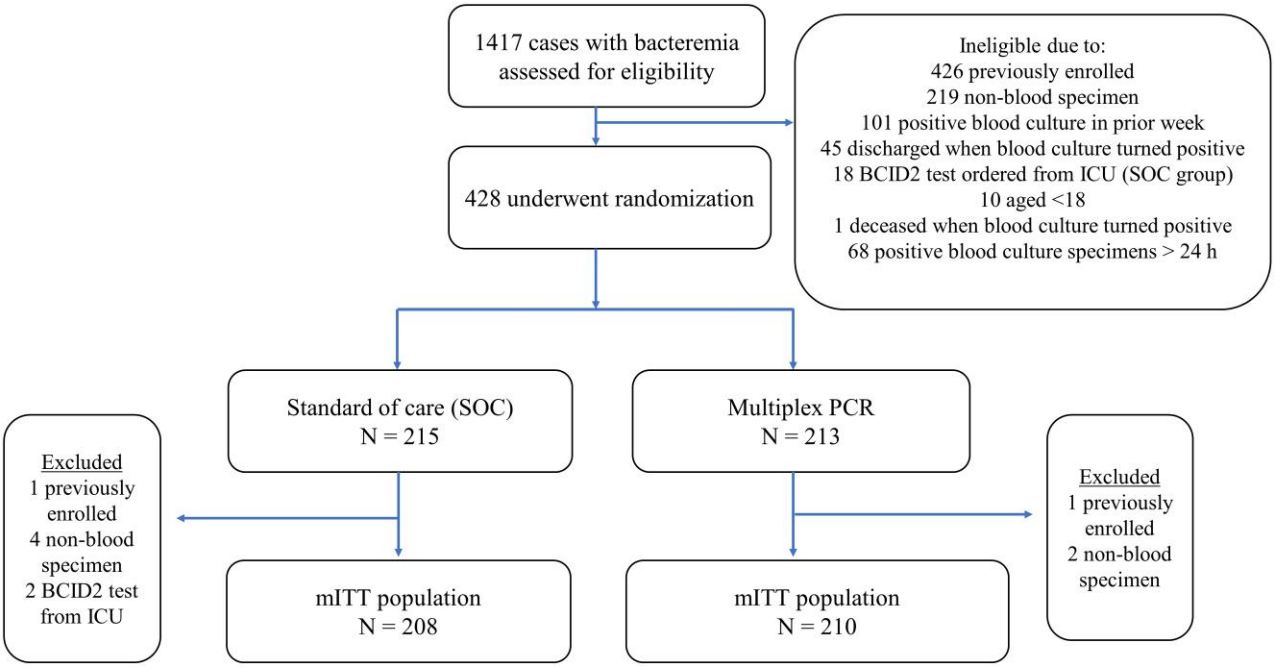


Figure 1. Flow diagram of participant eligibility, randomization, and inclusion. The mITT population included participants who met all the inclusion criteria and no exclusion criteria. Abbreviations: BCID2, BioFire BCID2 test; ICU, intensive care unit; Multiplex PCR, pathogen identification and antimicrobial resistance detection using the BioFire BCID2 test plus standard of care testing; mITT, modified intention-to-treat.

therapy within 120 hours are presented in Supplementary Table 1. The ID consultation rates did not differ between the study arms (Supplementary Table 2).

Blood culture contamination occurred in 12.5% (26/208) of the SOC patients and 17.1% (36/210) of the multiplex PCR patients ($P = .216$). Among these cases, antimicrobial therapy was not initiated in 30.8% (8/26) and 52.8% (19/36) of the patients in the SOC and multiplex PCR arms, respectively ($P = .120$) (Table 5; Supplementary Table 3).

Total antimicrobial use and duration were comparable between the arms (Supplementary Table 4). However, the multiplex PCR arm showed a numerical trend toward longer median effective antimicrobial durations (Supplementary Table 5) and shorter ineffective antimicrobial durations across most antimicrobial classes, except for third-generation cephalosporins (Supplementary Table 6), although these differences were not statistically significant.

Persistent bloodstream infections were less frequent in the multiplex PCR arm than in the SOC arm (5.7% vs 11.5%; $P = .038$) (Table 7). Among patients without blood culture clearance, carbapenem-resistant *K. pneumoniae* were only observed in the SOC arm (Supplementary Table 7). Source control interventions, including intravenous catheter removal, were more frequent in the multiplex PCR arm, whereas their timing was similar between the groups (Supplementary Table 8). The overall incidence of adverse events, including the acquisition of organisms with antimicrobial resistance, *C. difficile*, or candidemia, did not differ significantly between groups (28.4% in the SOC arm vs 22.4% in the multiplex PCR arm; $P = .178$) (Table 7). However, methicillin-resistant *S. aureus* (MRSA) acquisition was significantly lower in the multiplex arm (0.5% vs 2.9%; $P = .037$).

Thirty-day mortality did not differ between the study arms (14.4% in the SOC arm vs 13.8% in the multiplex PCR arm; $P = .889$), and other clinical outcomes, including length of stay, ICU admission rate, readmission rate, and healthcare costs, were comparable (Table 8).

Among 195 on-panel organisms in the multiplex PCR arm, 5 discrepancies (2.6%) were observed between BCID2 and SOC results: 4 species identification discrepancies and 1 resistance marker discrepancy (Supplementary Table 9).

The reduction in time to effective antimicrobial modification was observed only among patients randomized within 12 hours of blood culture positivity; other outcomes were generally comparable between the study arms (Supplementary Table 10).

DISCUSSION

In this prospective RCT of patients with bloodstream infections, integrating the BioFire BCID2 Panel into SOC testing reduced the median time to the first effective antimicrobial modification by 41.3 hours (23.2 vs 64.5 hours; $P < .001$).

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Table 1. Baseline Sociodemographic and Clinical Characteristics of Each Study Arm

Characteristics	Standard of Care (n = 208)	Multiplex PCR (SOC + BioFire BCID2) (n = 210)
Sociodemographics		
Male, no. (%)	113 (54.3)	117 (55.7)
Age, median [IQR]	71 [61, 80]	70 [60, 79]
Korean, no. (%)	207 (99.5)	207 (98.6)
Social deprivation index[a], high, no. (%)	84 (40.4)	78 (37.1)
Insurance type		
National health insurance	184 (88.5)	188 (89.5)
Medical Aid	24 (11.5)	22 (10.5)
Coexisting conditions		
Charlson comorbidity score, median [IQR]	2 [0, 2]	2 [1, 2]
Diabetes mellitus	85 (40.9)	82 (39.0)
Chronic heart disease	38 (18.3)	27 (13.8)
Chronic pulmonary disease	27 (13.0)	31 (14.8)
Chronic kidney disease	25 (12.0)	30 (14.3)
Acute kidney disease	23 (11.1)	16 (7.6)
Rheumatic disease	9 (4.3)	5 (2.4)
Solid tumor	67 (32.2)	77 (36.7)
Leukemia or lymphoma	23 (11.1)	35 (16.7)
Dementia	17 (8.2)	14 (6.7)
Stroke	33 (15.9)	31 (14.8)
Clinical characteristics at enrollment		
Length of hospital stay prior to randomization, median [IQR]	3 [1, 14.8]	4.5 [1, 15]
Pitt bacteremia score[b], median [IQR]	2 [0, 3]	2 [0, 2]
Admitted to ICU	62 (29.8)	61 (29.0)
Mechanical ventilation	20 (9.6)	20 (9.5)
Vasopressor use	83 (39.9)	76 (36.2)
Altered mental state	37 (17.8)	39 (18.6)
Hypotension[c]	92 (44.2)	96 (45.7)
Community-onset infection[d]	96 (46.2)	94 (44.8)
Source of bacteremia		
Intra-abdominal	59 (28.4)	57 (27.1)
Urinary	40 (19.2)	48 (22.9)
Respiratory	39 (18.8)	28 (13.3)
Catheter	25 (12)	40 (19)
Skin and soft tissue	12 (5.8)	7 (3.3)
Cardiovascular	9 (4.3)	4 (1.9)
Surgical site infection	3 (1.4)	4 (1.9)
Bone and joint	1 (0.5)	0 (0)
Unidentified	19 (9.1)	22 (10.5)

Abbreviations: ICU, intensive care unit; IQR, interquartile range; SOC, standard of care.

[a]A composite measure that reflects the level of socioeconomic disadvantage in a given geographic area, with higher values indicating greater deprivation.

[b]Pitt bacteremia score assesses the severity of acute illness and ranges from 0 to 14 points, with a score ≥4 commonly used to indicate critical illness and an increased risk of death.

[c]Defined as occurring within 24 h of randomization and meeting at least one of the following criteria: (a) a decrease in blood pressure of more than 30 mmHg (systolic) and 20 mmHg (diastolic), (b) the necessity of intravenous vasopressors, or (c) systolic blood pressure falling below 90 mmHg.

[d]Hospitalization duration ≤48 h prior to randomization.

Table 2. Pathogens and Antimicrobial Resistance Patterns in Each Study Arm

Microbial Characteristics	Standard of Care (n = 208), No. (%)	Multiplex PCR (n = 210), No. (%)
Gram-negative bacteria	121 (58.2)	104 (49.5)
Acinetobacter baumannii-calcoaceticus complex	6 (2.9)	6 (2.9)
Bacteroides fragilis	0 (0)	1 (0.5)
Enterobacterales	103 (49.5)	85 (40.5)
Enterobacter cloacae complex	6 (2.9)	4 (1.9)
Escherichia coli	50 (24)	49 (23.3)
Klebsiella aerogenes	3 (1.4)	4 (1.9)
Klebsiella oxytoca	2 (1)	1 (0.5)
Klebsiella pneumoniae group		
Klebsiella pneumoniae	26 (12.5)	16 (7.6)
Klebsiella variicola	2 (1)	1 (0.5)
Proteus spp.	0 (0)	5 (2.4)
Salmonella spp.	4 (1.9)	1 (0.5)
Serratia marcescens	2 (1)	1 (0.5)
Other *Enterobacterales*	8 (3.8)	3 (1.4)
Haemophilus influenzae	0 (0)	1 (0.5)
Neisseria meningitidis	0 (0)	0 (0)
Pseudomonas aeruginosa	2 (1)	4 (1.9)
Stenotrophomonas maltophilia	1 (0.5)	2 (1)
Gram-negative off panel pathogens[a]	9 (4.3)	5 (2.4)
Gram-positive bacteria	90 (43.3)	99 (47.1)
Enterococcus faecalis	9 (4.3)	9 (4.3)
Enterococcus faecium	8 (3.8)	14 (6.7)
Listeria monocytogenes	0 (0)	0 (0)
Staphylococcus aureus	9 (4.3)	9 (4.3)
Staphylococcus epidermidis	22 (10.6)	26 (12.4)
Staphylococcus lugdunensis	0 (0)	2 (1)
Other *Staphylococcus* spp.	18 (8.7)	19 (9)
Streptococcus agalactiae	0 (0)	1 (0.5)
Streptococcus pneumoniae	0 (0)	0 (0)
Streptococcus pyogenes	0 (0)	0 (0)
Other *Streptococcus* spp.	7 (3.4)	4 (1.9)
Gram-positive off-panel pathogens[a]	17 (8.2)	15 (7.1)
Yeasts	11 (5.3)	13 (6.2)
Candida albicans	4 (1.9)	6 (2.9)
Candida auris	0 (0)	2 (1)
Candida glabrata	1 (0.5)	0 (0)
Candida krusei	1 (0.5)	0 (0)
Candida parapsilosis	1 (0.5)	2 (1)
Candida tropicalis	2 (1)	1 (0.5)
Cryptococcus neoformans	0 (0)	1 (0.5)
Off-panel yeast pathogens[a]	2 (1)	1 (0.5)
Antimicrobial resistance		
Carbapenem-resistant *Enterobacterales*	11/103 (10.7)	5/85 (5.9)
ESBL-producing *Enterobacterales*	16/103 (15.5)	23/85 (27.1)
Methicillin-resistant *S. aureus*	4/9 (44.4)	5/9 (55.6)
Methicillin-resistant	31/40 (77.5)	37/46 (80.4)

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Table 2. Continued

Microbial Characteristics	Standard of Care (n = 208), No. (%)	Multiplex PCR (n = 210), No. (%)
staphylococci other than *S. aureus*		
Vancomycin-resistant enterococci	5/19 (26.3)	7/25 (28)
Multidrug-resistant[b] *Pseudomonas* spp.	1/2 (50)	1/4 (25)
Multidrug-resistant[b] *Acinetobacter* spp.	3/9 (33.3)	3/9 (33.3)
Multidrug-resistant[c] *Candida* spp.	0/10 (0)	0/12 (0)

Microorganisms listed individually in the table, including those reported under group-level targets such as *Enterobacterales*, other *Staphylococcus* spp., and other *Streptococcus* spp., were classified as on-panel organisms.

Abbreviation: ESBL, extended-spectrum β-lactamase.

[a]Defined as microorganisms not included in the BioFire BCID2 Panel.

[b]Defined as non-susceptible to at least one agent in three or more antimicrobial categories.

[c]Defined as an isolate not susceptible to ≥1 agent in ≥2 drug classes.

Table 3. Comparison of Time to Randomization and Microbiological Results in Each Study Arm

Time Interval	Standard of Care (n = 208)	Multiplex PCR (n = 210)	*P* Value
Time from BC positivity to randomization, h, median [IQR]	5.0 [1.0, 10.3]	4.2 [0.9, 9.7]	0.566
Time from randomization to BCID2 result, h, median [IQR]	NA	1.9 [1.6, 2.6]	NA
Time from randomization to MALDI-TOF MS result, h, median [IQR]	24.4 [22.5, 26.2]	24.3 [22.7, 25.8]	0.876
Time from randomization to phenotypic AST result, h, median [IQR]	51.6 [50.5, 73.1]	51.1 [50.6, 73.2]	0.739

Abbreviations: AST, antimicrobial susceptibility testing; BC, blood culture; IQR, interquartile range; NA, non-applicable.

This reduction is primarily attributable to the rapid transition from empirical to targeted therapy guided by simultaneous pathogen identification and resistance marker detection using the multiplex PCR panel. Similarly, the RAPIDS-GN trial, which evaluated the Accelerate Pheno system (Accelerate Diagnostics, Inc., Tucson, AZ, United States) for rapid phenotypic identification and susceptibility testing, demonstrated faster antimicrobial modification in the rapid diagnostic arm [5]. In our study, the reduction was consistently observed across most subgroups, including various pathogen types, ICU statuses, and laboratory staffing hours. In RAPIDS-GN, this benefit was also observed in non-neutropenic patients and those with a Pitt bacteremia score ≥2, but not in the other prespecified subgroups [5]. Direct-from-blood diagnostic approaches may further shorten the time to actionable microbiological results by eliminating the need for blood culture growth

Table 4. Time to First Effective Antimicrobial Modification by the Study Arm Across Subgroups

Group	Hours From Randomization to First Effective Antimicrobial Modification: Standard of Care (n = 208), Median [IQR]	Multiplex PCR (n = 210), Median [IQR]	Difference	*P* Value[a]
All	64.5 [25.9, 120]	23.2 [3.9, 119.5]	41.3	<.001
Regular lab hours[b]	66.7 [27.4, 120]	9.3 [3.1, 92.2]	57.4	<.001
Off-hours[c]	60.5 [25.1, 120]	37.5 [5.8, 120]	23.0	<.001
Gram-negative bacteria[d]	77.1 [29.3, 120]	30.5 [5.7, 81.5]	46.6	<.001
Gram-positive bacteria[e]	53.5 [21.1, 120]	22.7 [3.9, 120]	30.8	.001
Yeasts[f]	27.7 [24.7, 57.6]	2.9 [2.2, 6.6]	24.8	<.001
In ICU at randomization[g]	84.8 [30.2, 120]	20.5 [4.3, 82.1]	64.3	<.001
Not in ICU at randomization[h]	55.5 [24.7, 120]	23.4 [3.8, 120]	32.1	<.001
Resistant organisms[i]	50.8 [13.5, 120]	23.5 [3.9, 120]	27.3	.037
Susceptible organisms[j]	72.4 [28.9, 120]	17.8 [4.1, 100.9]	54.6	<.001
Modification[k]	32.7 [15.1, 69.1]	8.0 [3.0, 33.8]	24.7	<.001
Escalation[l]	26.1 [9.9, 50.6]	6.1 [2.3, 19.1]	20.0	<.001
De-escalation[m]	60.5 [31.3, 91.9]	27.6 [8.4, 70.7]	32.9	<.001
On-panel organisms[n]	59.0 [24.8, 120]	13.1 [3.8, 81.1]	45.9	<.001
Off-panel organisms[o]	100.1 [30.7, 120]	120 [25.2, 120]	−19.9	.921

Abbreviations: ICU, intensive care unit; IQR, interquartile range.

[a]Medians were compared using the Mann–Whitney U test after the Shapiro–Wilk normality test.

[b]Patients with blood cultures flagged positive during regular laboratory hours (Mon–Sat, 7:00–16:00). SOC, n = 71; multiplex PCR (SOC + BioFire BCID2), n = 79.

[c]Patients with blood cultures flagged positive during non-regular laboratory hours. SOC, n = 137; multiplex PCR, n = 131.

[d]Patients with blood cultures positive for Gram-negative bacteria. SOC, n = 117; multiplex PCR, n = 104.

[e]Patients with blood cultures positive for Gram-positive bacteria. SOC, n = 86; multiplex PCR, n = 96.

[f]Patients with blood cultures positive for yeast SOC, n = 11; multiplex PCR, n = 13.

[g]SOC, n = 62; multiplex PCR, n = 61.

[h]SOC, n = 146; multiplex PCR, n = 149.

[i]Patients with blood cultures positive for organisms meeting predefined antimicrobial resistance criteria (including carbapenem-resistant *Enterobacterales*, oxacillin-resistant *Staphylococcus* spp., and other predefined multidrug-resistant organisms), as described in the Methods. SOC, n = 71; multiplex PCR, n = 81.

[j]Patients with blood cultures positive for organisms without antimicrobial resistance. SOC, n = 137; multiplex PCR, n = 129.

[k]Patients who underwent antimicrobial modification (escalation, de-escalation, or lateral change), excluding those with no change. SOC, n = 142; multiplex PCR, n = 156.

[l]Patients who underwent antimicrobial escalation. SOC, n = 88; multiplex PCR, n = 108.

[m]Patients who underwent antimicrobial de-escalation. SOC, n = 45; multiplex PCR, n = 38.

[n]Patients with at least one BCID2 on-panel organism, including polymicrobial cultures with both on-panel and off-panel organisms. SOC, n = 182; multiplex PCR, n = 188.

[o]Patients with only BCID2 off-panel organisms, including false-positive blood culture results. SOC, n = 26; multiplex PCR, n = 22.

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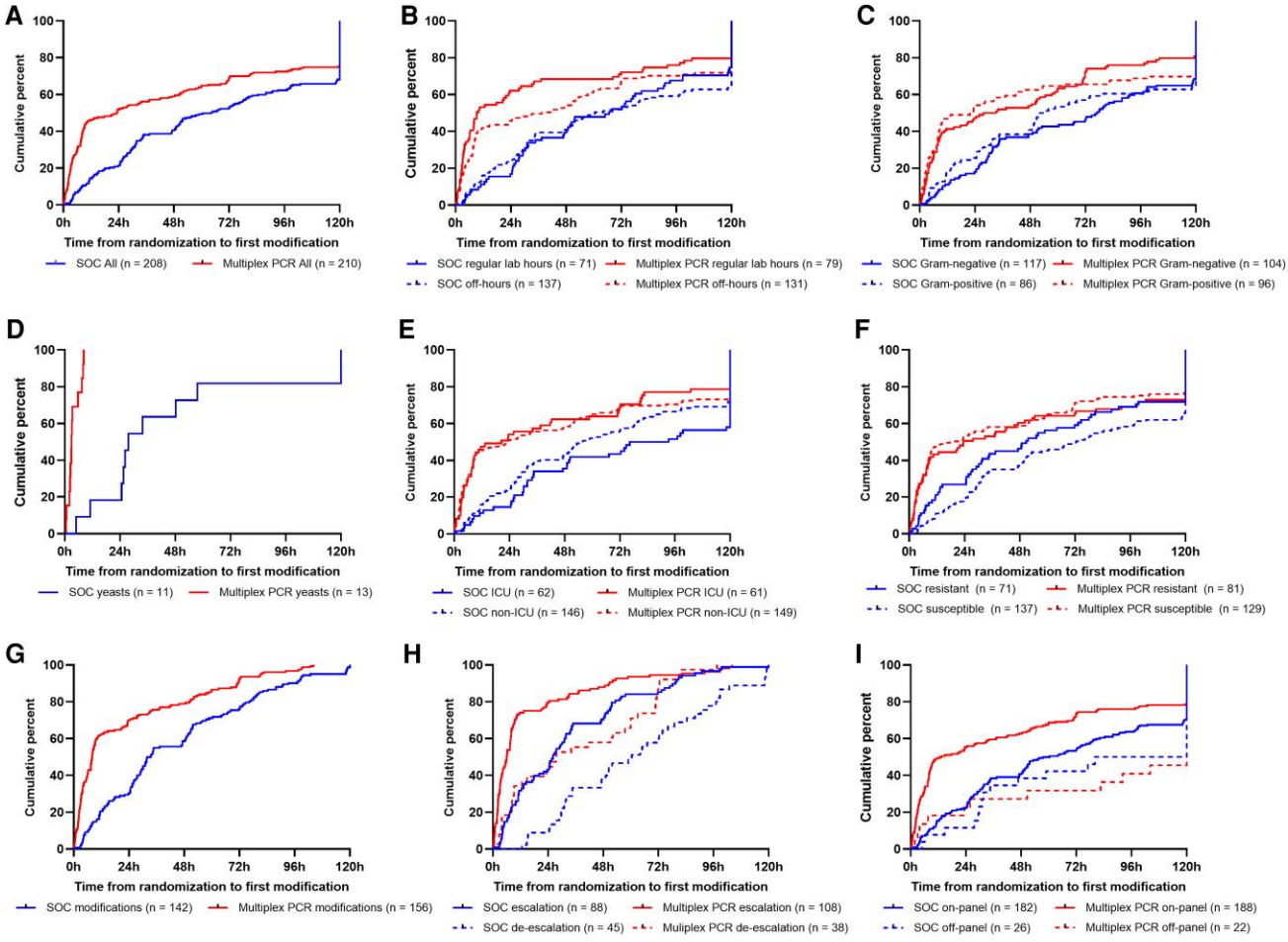


Figure 2. Time to first effective antimicrobial modification in the mITT population and key clinical subgroups based on the treatment arm. *A*, Total population. Cumulative incidence of time from randomization to the first effective antimicrobial modification. *B*, Laboratory operating hours. Comparisons between patients with blood cultures flagged positive during regular hours (Mon–Sat, 07:00–16:00) and off-hours. *C*, Pathogen type (bacteria). Comparisons between Gram-negative and Gram-positive bacterial infections. *D*, By pathogen type (yeast). *E*, By admission ward. Comparisons between patients admitted to the ICU and non-ICU wards at the time of randomization. *F*, By antimicrobial resistance. Comparisons between resistant organisms and susceptible organisms. *G*, Patients with antimicrobial modification, including lateral change. *H*, By type of modification. Comparison between antimicrobial escalation and de-escalation. *I*, By BCID2 panel coverage. On-panel organisms and off-panel organisms. Abbreviations: SOC, standard of care; Multiplex PCR, pathogen identification and antimicrobial resistance detection using the BioFire BCID2 test plus SOC; h, hour; ICU, intensive care unit.

[17]. However, a recent RCT using T2Bacteria panel (T2Biosystems, Lexington, MA) showed that providing rapid bacterial results did not shorten the duration of broad-spectrum antibiotic therapy [18].

The incidence of persistent bloodstream infection was lower in the multiplex PCR arm than in the SOC arm (Table 7), a finding not observed in previous diagnostic trials. Although earlier antimicrobial modification and a higher proportion of effective antimicrobial therapy during the early phase of infection were observed in the multiplex PCR arm (Tables 4 and 6), effective antimicrobial therapy at 120 hours was not associated with persistent bloodstream infection (Supplementary Table 11). Moreover, source control interventions, including intravenous catheter removal, were more frequent in the multiplex PCR arm (Supplementary Table 8). Thus, multiple factors may have contributed to the lower incidence of persistent bloodstream infection, which warrants further study.

The multiplex PCR arm showed a higher proportion of cases in which antibiotics were not initiated for contaminants compared with the SOC arm (52.8% vs 30.8%; $P = .120$) (Table 5; Supplementary Table 3). This finding is consistent with that of a previous study showing that rapid diagnostic testing may improve judicious antimicrobial use [2].

Antimicrobial exposure can alter microbiota composition and diversity, facilitating the emergence of resistant organisms and weakening colonization resistance [19]. Although the composite outcome of acquiring antimicrobial-resistant organisms, *C. difficile*, or candidemia did not differ significantly between the groups, the multiplex PCR arm showed a significantly lower incidence of MRSA acquisition and a trend toward reduced

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Table 5. Antimicrobial Therapy Modifications in Each Study Arm

Antimicrobial Modification	Standard of Care (n = 208)	Multiplex PCR (n = 210)	*P* Value
Escalation	88 (42.3)	108 (51.4)	.272
Switch to broader therapy	29 (13.9)	34 (16.2)	
Addition of one or more therapy	54 (26.0)	72 (34.3)	
Dose increase	5 (2.4)	2 (1.0)	
De-escalation	45 (21.6)	38 (18.1)	
Switch to narrower therapy	24 (11.5)	20 (9.5)	
Discontinuation of one or more therapy	16 (7.7)	15 (7.1)	
Dose reduction	3 (1.4)	3 (1.4)	
IV to PO therapy	2 (1.0)	0 (0)	
Lateral change[a]	9[b] (4.3)	10[c] (4.8)	
No antimicrobial modification	66 (31.7)	54 (25.7)	
Empirical therapy continued	38 (18.3)	25 (11.9)	
No therapy due to contamination[d]	8 (3.8)	19 (9.0)	
No therapy due to false positive[e]	1 (0.5)	3 (1.4)	
Discharged before modification	0 (0)	1 (0.5)	
No effective[f] therapy	19 (9.1)	6 (2.9)	

Data are presented as no. (%) unless otherwise specified.

[a]Changes not classified as escalation or de-escalation, mainly reflecting movement within the same rank.

[b]Lateral changes in the standard of care group included meropenem to ceftazidime/avibactam (n = 2), piperacillin/tazobactam to ertapenem (n = 3), ertapenem to cefepime (n = 1), cefazolin to ampicillin (n = 1), piperacillin/tazobactam to cefepime (n = 1), and vancomycin to teicoplanin (n = 1).

[c]Lateral changes in the multiplex PCR group included meropenem to ceftazidime/avibactam (n = 2), piperacillin/tazobactam to ertapenem (n = 4), piperacillin/tazobactam to cefepime (n = 2), ertapenem to piperacillin/tazobactam (n = 1), and micafungin to anidulafungin (n = 1).

[d]No antibiotic therapy was initiated in 30.8% (8/26) of SOC and 52.8% (19/36) of multiplex PCR contamination cases, representing all identified contaminants in each group (Fisher's exact test, *P* = .120).

[e]False-positive results were defined as positive blood culture signals without microbiological evidence of viable organisms (no organisms on Gram stain, no growth on subculture, and no recovery on repeat culture).

[f]Effective antimicrobial therapy was defined based on in vitro susceptibility results.

Table 6. Proportion of Patients Receiving Effective[a] Antimicrobial Therapy in Each Study Arm

Time From Randomization	Standard of Care, No. (%) (n = 208)	Multiplex PCR, No. (%) (n = 210)	*P* Value
0 h	104 (50.0)	92 (43.8)	.240
12 h	137 (65.9)	181 (86.2)	<.001
24 h	141 (67.8)	184 (87.6)	<.001
48 h	167 (80.3)	192 (91.4)	.001
72 h	177 (85.1)	198 (94.3)	.002
96 h	186 (89.4)	200 (95.2)	.028
120 h	189 (90.9)	204 (97.1)	.007

[a]Effective antimicrobial therapy was based on in vitro susceptibility results and included patients receiving effective therapy, regardless of whether antimicrobial therapy was modified.

Table 7. Microbiological Outcomes of the Study Arms

Outcomes	Standard of Care, No. (%) (n = 208)	Multiplex PCR, No. (%) (n = 210)	*P* Value
Persistent bloodstream infection[a]	24 (11.5)	12 (5.7)	.038
Acquisition of antimicrobial-resistant organisms[b], *C. difficile*[c], or candidemia[d]	59 (28.4)	47 (22.4)	.178
CRE	28 (13.5)	25 (11.9)	.661
KPC producers	21 (10.1)	21 (10)	
NDM producer	1 (0.5)	1 (0.5)	
Non-carbapenemase producers	6 (2.9)	3 (1.4)	
ESBL-producing *Enterobacterales*	10 (4.8)	7 (3.3)	.470
MDR[e] *Pseudomonas* spp.	10 (4.8)	5 (2.4)	.200
MDR[e] *Acinetobacter* spp.	9 (4.3)	9 (4.3)	>.999
VRE	9 (4.3)	11 (5.2)	.820
MRSA	7 (2.9)	1 (.5)	.037
Healthcare facility-onset *C. difficile*	5 (2.4)	5 (2.4)	>.999
Candidemia	13 (6.3)	5 (2.4)	.057

Abbreviations: CRE, carbapenem-resistant *Enterobacterales*; ESBL, extended-spectrum beta-lactamase; MDR, multidrug-resistant; MRSA, methicillin-resistant *Staphylococcus aureus*; VRE, vancomycin-resistant *Enterococcus* spp.

[a]Isolation of the original pathogen from follow-up blood cultures ≥3 d (bacteria) or ≥5 d (yeast) post-randomization.

[b]Detection in any clinical specimen collected post-randomization and on/after hospital day 4, within 30 d of randomization, provided no prior positive culture for the same organism in the preceding 3 m (including unknown status). Antimicrobial-resistant organisms included carbapenem-resistant *Enterobacterales*, oxacillin-resistant *Staphylococcus* spp., and other predefined multidrug-resistant organisms, as described in the Methods.

[c]A positive result for *C. difficile* toxin after randomization and on/after day 4 of hospitalization.

[d]Defined as first-time isolation post-randomization or emergence of a different *Candida* species in patients with preexisting candidemia.

[e]Defined as non-susceptibility to at least one agent in three or more antimicrobial categories.

candidemia (Table 7). Similarly, the RAPIDS-GN trial reported a reduction in CRE acquisition (0% [0/222] vs 2% [5/226], *P* = .06) [5]. However, a similar reduction was not observed in the multinational FAST trial evaluating rapid phenotypic AST using the Reveal system (bioMérieux) [7]. Given the clinical burden associated with antimicrobial-resistant infections [20], *C. difficile* [21], and candidemia [22], the potential impact of rapid diagnostic testing on secondary infections merits further evaluation.

Notably, a relatively small number of yeast pathogens were detected in both groups (n = 11 in the SOC arm and n = 13 in the multiplex PCR arm) (Table 4). Except for a single isolate of *Saccharomyces cerevisiae* in the SOC arm and *Cryptococcus neoformans* in the multiplex PCR arm, all isolates were *Candida* species. Despite the limited number of events, a clear separation was observed in the Kaplan–Meier curve (Figure 2*D*). This finding may be partly explained by the differences in antifungal management. Unlike antibacterial agents, which are often initiated empirically before microbiological confirmation, antifungal therapy

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Table 8. Clinical and Economic Outcomes for Each Study Arm

Outcomes	Standard of Care (n = 208)	Multiplex PCR (n = 210)	*P* Value
Clinical outcomes			
30-d mortality, no. (%)	30 (14.4)	29 (13.8)	.889
Discharge within 30 d after randomization, no. (%)	132 (63.5)	141 (67.1)	.472
Readmission within 30 d after discharge, no. (%)	14 (6.7)	18 (8.6)	.582
Healthcare utilization			
Length of hospital stay up to 30 d, median [IQR]	13 [7, 28]	14 [8, 25]	.739
Length of ICU stay[a], days, median [IQR]	5.5 [2, 12.3]	6 [2, 13]	.585
ICU admission[b], no. (%)	16 (11.0)	9 (6.0)	.147
Costs			
Total Hospital cost, median [IQR]	$9904 ($4185, $21 206)	$9650 ($5080, $22 004)	.624
Laboratory cost, median [IQR]	$832 ($415, $2112)	$1015 ($586, $2157)	.051
Antimicrobial drug cost, median [IQR]	$238 ($90, $960)	$270 ($118, $993)	.334

Abbreviations: ICU, intensive care unit; IQR, interquartile range.

[a]Calculated among patients in the ICU at randomization. Standard of care, n = 62; multiplex PCR, n = 61.

[b]Assessed among patients not in the ICU at randomization. Standard of care, n = 146; multiplex PCR, n = 149.

is rarely initiated for suspected bloodstream infections without microbiological evidence of yeast or *Candida* species. Therefore, the shorter time to effective antimicrobial modification primarily reflected the earlier introduction of antifungal therapy, guided by the microbiological results in the multiplex PCR arm (Table 4). Persistent candidemia is relatively common (25.1% of patients) despite available antifungal therapies and is associated with poor outcomes [23, 24]. In the present study, the rate of persistent candidemia was lower in the multiplex PCR arm than in the SOC arm (1/12 [8.3%] vs 4/10 [40%]; Fisher's exact test, $P = .135$) (Table 2; Supplementary Table 7). This reduction may suggest a potential benefit of earlier antifungal optimization in reducing persistent infections. The lack of significant differences observed in 30-day mortality, length of stay, and hospital costs aligns with the findings of previous RCTs [2, 5, 7], where the impact of rapid diagnostics on clinical outcomes was attenuated by the clinical complexity and baseline comorbidities of patients with bloodstream infections [5]. Similarly, the FAST trial found no superiority of rapid phenotypic AST over standard testing in the 30-day desirability of outcome ranking, a composite outcome integrating efficacy and safety [7]. In our study, most patients had an identifiable source of bacteremia, which may have further confounded the assessment of clinical outcomes (Table 1). Moreover, approximately half of patients had already received effective empirical therapy at enrollment (50.0% in the SOC arm and 43.8% in the multiplex PCR arm) (Table 6), potentially limiting the clinical benefit of rapid diagnostics. The lower proportion in the multiplex PCR arm, however, may have allowed greater opportunity for treatment modification and should be considered when interpreting clinical outcomes.

A key strength of this study is its pragmatic, randomized design, reflecting real-world implementation of the BCID2 panel within routine clinical workflows. Both study arms were managed under the same AS program, and the only study-specific intervention was text notification of randomization and initial blood culture identification results. Thus, the study evaluated the impact of rapid diagnostic information without research-driven stewardship interventions. Notably, a 41.3-hour reduction in time to effective antimicrobial modification was achieved despite limited off-hour processing, highlighting the feasibility of integrating rapid diagnostics into routine laboratory workflows under real-world staffing constraints [25]. This supports the feasibility of integrating rapid diagnostics into routine laboratory workflows. In Korea, reimbursement for BCID2 testing is currently limited to positive blood cultures from ICU patients within 8 hours of positivity. Following this study, our institution implemented 24/7 BCID2 testing for ICU patients based on provider requests, while eligible ICU blood cultures undergo laboratory-initiated reflex testing during regular laboratory hours.

Despite its strengths, this study had several limitations. First, the generalizability of our findings may be limited by institution-specific laboratory workflows. Organism identification in the SOC arm relied on MALDI-TOF MS after overnight subculture, and both conventional and molecular testing were performed only during regular laboratory hours. As laboratory workflows and testing availability vary across institutions, the clinical impact of rapid molecular testing should be interpreted in the context of local practice. Second, our study was not powered to detect differences in 30-day mortality or length of hospital stay. These outcomes are influenced by factors beyond antimicrobial timing, including concurrent infections, source control, and comorbidities. Mortality benefits may become more apparent when rapid diagnostics are combined with AS in larger populations [8]. Third, specific clinical characteristics and management practices in our cohort may limit generalizability. The overall prevalence of *S. aureus* bacteremia was 4.3% (18/418), which is lower than the rates reported in a previous trial (8.6%) [2] and review literature (15.4%–20.7%) [26]. In addition, formal ID consultation was performed in 261/418 patients (62.4%) overall (Supplementary Table 2), higher than rates reported in previous studies of Gram-negative bloodstream infections (36.5%) [27], candidemia (45.9%) [28], and *S. aureus* bacteremia (49.5%) [29]. Given the association between ID consultation and improved outcomes in these infections [27–29], the benefit of rapid diagnostics may differ in settings where ID consultation is less frequently available. Finally, a small potential for selection bias exists due to the post-randomization exclusion of 2 patients in the SOC arm

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whose treating physicians subsequently ordered routine clinical BCID2 testing in the ICU.

Rapid molecular diagnostics combined with AS program support significantly reduces the time to the first effective antimicrobial modification. These findings support the integration of rapid diagnostics with established AS program for the optimal management of bloodstream infections.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author contributions. K. J. K. conceptualized and designed the study, served as the principal investigator, secured funding, collected and analyzed the data, and drafted the manuscript. Y. K. Y. contributed to the conceptualization and collected and analyzed the data. J. Y. K., M. J. K., J. W. S., and M. H. N. collected and analyzed the data; J. W. S. and M. H. N. supervised the study. All authors reviewed and edited the final version.

Data availability. Data are not publicly available.

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